UNIT I

(History of Pharmacopoeia, Impurities, Limit tests)

SECTION A (2 Marks)

1. What is Pharmacopoeia? Enlist different Pharmacopoeias.
2. What are limit tests? Why are they performed?
3. What is the principle involved in limit test of chloride?
4. What is the modified limit test for chloride?
5. Name the apparatus used in arsenic limit test.
6. What are the contents of a monograph?
7. Write the molecular formula of white vitriol.
8. What do you mean by impurity?

SECTION B (10 Marks)

1. Define impurity and discuss various sources of impurities in pharmaceutical substances.
2. Define limit test. List various limit tests and discuss limit test for sulphate.
3. Discuss in detail the limit test for arsenic with apparatus and principle.
4. Describe the assay of heavy metals.

SECTION C (7 Marks)

1. Discuss history of Pharmacopoeia.
2. Explain principle, reaction and procedure of limit test for arsenic.
3. Discuss the limit test of chloride.
4. Write detailed note on sources and types of impurities.

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UNIT II

(Buffers, Electrolytes & Dental Products)

SECTION A (2 Marks)

1. State ideal properties of buffer solutions.
2. What is isotonic solution and iso-osmotic solution?
3. What are extra and intracellular electrolytes? Write their functions.
4. Give uses of potassium chloride.
5. Write WHO composition of ORS.
6. What is achlorhydria?
7. Give function of ORS.
8. Define dentifrices with examples.
9. What is zinc eugenol cement?

SECTION B (10 Marks)

1. What are acidifiers? Describe preparation, properties, uses and assay of ammonium chloride.
2. Discuss electrolytes used in replacement therapy.
3. Explain buffer equations and buffer capacity.
4. Describe preparation and uses of ORS.

SECTION C (7 Marks)

1. Discuss methods of measurement of tonicity.
2. What are different methods of calculation of isotonicity?
3. Explain physiological role of sodium, potassium and calcium.
4. Outline anticaries agents and explain role of fluoride in dental caries.
5. Write a short note on physiological acid-base balance.

UNIT III

(Gastrointestinal Agents & Antimicrobials)

SECTION A (2 Marks)

1. Define cathartics with examples.
2. Define acidifying agents.
3. What is antacid?
4. Write uses of sodium bicarbonate.
5. Classify inorganic antimicrobial agents.
6. List examples of iodine and its preparations.
7. Explain role of kaolin and bentonite.

SECTION B (10 Marks)

1. What are antacids? Describe ideal properties and uses of antacids.
2. Describe preparation, properties, assay and uses of sodium bicarbonate.
3. Describe preparation, properties, assay and uses of hydrogen peroxide.
4. What are antimicrobials? Explain mechanism of action.

SECTION C (5 / 7 Marks)

1. Explain mechanism of action and uses of hydrogen peroxide.
2. Write properties, storage and uses of potassium permanganate and boric acid.
3. Classify cathartics according to mechanism of action with examples.
4. Write preparation, properties and uses of magnesium sulphate.
5. Write detailed note on iodine and its preparations.

UNIT IV

(Miscellaneous Compounds)

SECTION A (2 Marks)

1. Define expectorants with examples.
2. What are hematinics?
3. Define emetics with examples.
4. What are astringents? Give examples.
5. Paraphrase role of activated charcoal.
6. Enlist uses of poison and antidote.
7. What are the uses of potash alum?

SECTION B (10 Marks)

1. What are emetics? Describe preparation, properties, uses and assay of copper sulphate.
2. Write detailed note on poison and antidote. Explain sodium thiosulphate.
3. What are hematinics? Explain preparation, properties and uses of ferrous sulphate.
4. Write preparation and uses of ammonium chloride and potassium iodide as expectorants.

SECTION C (7 Marks)

1. Explain cyanide poisoning and its treatment.
2. Write preparation, assay and uses of sodium thiosulphate.
3. Explain preparation, properties and uses of ferrous sulphate.
4. Write short note on activated charcoal.

UNIT V

(RADIOACTIVITY)

SECTION A (2 Marks)

1. What is radioactivity? Give its unit.
2. What is half-life of radioactive elements?
3. Differentiate between α, β and γ radiations.

SECTION C (7 Marks)

1. Write precautions during handling and storage of radioactive substances.
2. Discuss pharmaceutical applications of radioactive substances.
3. Explain measurement of radioactivity.